SE LAB 3 SUBMISSION - Component Modelling & Architectural Pattern Selection

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Architecture Selection

We chose Microservices Architecture for the Internal Microservice Catalog and Health Portal System.

Architectural Choice and Justification

Microservices Architecture is the most suitable choice for this system because the portal contains multiple distinct functionalities such as service registration and catalog management, health monitoring, dependency graph visualization, authentication, and downtime alerting. These functionalities can be implemented as separate services with well-defined interfaces.

Reason 1: Independent scaling and deployment
The system performs continuous health checks on registered microservices at 30-second intervals and must support an interactive dependency graph containing up to 200 services. With a microservices architecture, components such as the Health Monitoring Service and Dependency Graph Service can be scaled independently according to their workload without requiring the entire system to be scaled. Individual services can also be updated or deployed independently. This is suitable for a system where monitoring and catalog workloads may vary independently.

Reason 2: Fault isolation
The system must continuously monitor microservice availability and detect failures after consecutive unsuccessful health probes. Microservices architecture provides fault isolation by separating major functionalities into independent services. A failure in one service, such as the Dependency Graph Service or Catalog Service, does not necessarily require the entire portal to stop functioning. This supports the system's requirement for continuous health monitoring and timely detection of service failures.

Security Advantage

Security is addressed through a dedicated Authentication and Access Control Service. The portal can require users to authenticate before accessing protected microservice catalog and health-monitoring functions. This directly supports the requirement that unauthenticated users must be denied access to protected microservice information and monitoring functions.

Performance Benefit

Microservices architecture provides independent scaling, allowing resource-intensive operations such as frequent health checks and dependency graph processing to be scaled separately. This helps the system maintain performance while monitoring multiple registered microservices and rendering dependency graphs containing up to 200 services.

COMPONENT DIAGRAM-

